SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Coding & Solution

Date	29 June 2026
Team ID	N/A
Project Name	OptiCrop Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/pavan09876519/OPTICROP.git
Programming Language(s)	Python, HTML,CSS
Framework(s) Used	Streamlit

Key Features Implemented	• Crop recommendation system • User-friendly web interface • Agricultural production optimization
Pending / Incomplete Features	• Database integration • User login/authentication • Cloud deployment
Setup / Run Instructions	• Install Python and required libraries. • Install Streamlit (pip install streamlit). • Open the project folder in VS Code. • Run the command: • streamlit run app.py • Open the local URL shown in the browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments: The project is developed using Python, HTML, CSS, and Streamlit with a modular structure, readable code, and version control through GitHub. The solution successfully addresses the core objective of optimizing agricultural production by providing crop recommendations through an interactive web application.